Multimodal Link Prediction Using Graph-Enhanced Text-Visual Representations for Knowledge Graphs

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Abstract—To address the limitations of existing art knowledge graphs in visual modality coverage, semantic label completeness, relation expressiveness, and data scale, this paper constructs ARTNET, an interconnected multimodal infrastructure for the art domain. Built on large-scale art image data, the framework integrates structural information, textual semantics, and visual features, and further establishes multimodal link prediction and edge-type scoring tasks together with their evaluation protocols, thereby enabling more systematic, reproducible, and fine-grained evaluation. Experimental results show that ARTNET provides a relatively complete and sustainable foundation for art object understanding, relation reasoning, and standardized research in this domain, while also supporting future multimodal benchmark construction.

Index Terms—Knowledge graphs; artworks; multimodal learning; link prediction; knowledge graph embedding; object understanding; ARTNE

1 Introduction

Knowledge graph (KG) is a semantic network that organizes entities, concepts, and their relationships in a structured manner, providing a unified data representation for tasks such as knowledge organization, semantic retrieval, relational reasoning, and intelligent decision making [1]. Conventional knowledge graphs typically model knowledge in the form of triplets (h, r, t), where a head entity, a relation, and a tail entity jointly describe real-world objects and their associations [7]. However, existing research shows that KG methods primarily focus on the symbolic modeling of entities, attributes, and relations, with an emphasis on graph connectivity and the inferential value of semantic relations, while paying relatively limited attention to the visual content carried by entities [32]. This modeling paradigm is effective for domains where textual and structured attributes dominate [33]; however, in the art domain, relying solely on symbolic relations and textual descriptions is often insufficient to capture the full semantic richness of knowledge objects [34].

Knowledge representation for artworks differs substantially from that of general entities [12], [35]. Artistic knowledge is reflected not only in structured attributes such as creator, period, and material, but also in the visual content and cultural meanings of artworks [12], [36]. Relying only on textual fields or graph-structured relations usually captures external attributes of artworks, but fails to adequately represent their image content, visual style, cultural symbolism, and art-historical associations [36]. Therefore, artistic knowledge is not merely a collection of structural relations, but a composite knowledge system formed jointly by graph structure, textual semantics, and visual content [36]. This indicates that knowledge representation in the art domain requires not only the organization of entity-relation data, but also the integration of structural, textual, and visual information within a unified framework, so as to support multimodal understanding, relational reasoning, and experimental evaluation [12], [36].

Although knowledge graphs have received increasing attention in cultural heritage, digital humanities, and art computing [37], [38], existing art knowledge graphs still show three main limitations. First, they place greater emphasis on metadata than on visual content, which makes it difficult to represent the key visual semantics within artworks [39]. Second, their label systems and relation modeling remain incomplete, which limits their ability to support fine-grained comparison, relation inference, and cross-modal alignment [40], [41]. Third, their experimental foundations remain weak, as they lack unified data organization, task definitions, and evaluation protocols, making it difficult to support reusable and comparable research [31]. Overall, existing art knowledge graphs have not yet formed a systematic research infrastructure that can jointly support multimodal representation, task modeling, and standardized evaluation.

The above issues indicate that the key challenge in knowledge modeling for the art domain lies not only in data scarcity or insufficient relations, but also in the strongly multimodal nature of artistic knowledge itself [42]. Unlike general knowledge graphs, which mainly rely on textual descriptions and symbolic relations, the core semantics of artistic knowledge are often distributed across the interaction between visual and linguistic modalities. A more complete knowledge representation therefore requires joint modeling of image content, textual descriptions, structural relations, and label systems [43]. At the same time, this multimodal nature increases the complexity of data construction and places higher demands on task design and evaluation mechanisms. At the data level, it is necessary to address the alignment among text, images, and labels, as well as issues such as missing modalities, uneven quality, and inconsistent granularity. At the task level, beyond classical knowledge graph tasks, it is also necessary to introduce tasks oriented toward visual understanding and cross-modal semantic modeling. However, without a unified data organization process, clear task definitions, and standardized evaluation protocols, it remains difficult to compare different studies effectively [31]. Therefore, what the art domain urgently needs is not simply a larger isolated graph, but a research infrastructure that can integrate multimodal data, support multiple types of research tasks, and provide a unified evaluation mechanism.

Therefore, this paper proposes ARTNET to establish an interconnected multimodal knowledge infrastructure for the art domain. ARTNET does not aim simply to increase the number of entities or relations in conventional art knowledge graphs. Instead, from the perspective of research infrastructure, it builds a multimodal platform that organizes graph structures, textual information, visual information, and label systems within a unified framework, thereby supporting task definition, model comparison, and evaluation protocols in art-oriented scenarios. The scope of this study is organized around three levels. At the data level, ARTNET constructs a scalable art knowledge base that connects and structurally represents entities, relations, text, and images, in order to address the limited multimodal support, incomplete relation coverage, and restricted scale of existing art knowledge graphs [40], [41]. At the task level, this paper focuses on relation modeling for artistic knowledge and defines multimodal tasks for art-oriented scenarios, including multimodal link prediction and a multimodal knowledge graph embedding task for edge-type scoring. These tasks evaluate how well models understand association patterns among artistic entities and the semantics of their relations. At the evaluation level, ARTNET provides a relatively consistent experimental setting, data split strategy, and set of evaluation metrics, offering a reusable and comparable benchmark framework for future research. It should be emphasized that this paper does not focus on achieving locally optimal performance on a single visual task. Rather, it aims to build an infrastructure that can continuously support object understanding, relational reasoning, and multimodal learning research. In this sense, ARTNET is positioned as a multimodal knowledge graph research framework for the art domain rather than a single algorithmic system.

The main contributions of this paper are summarized as follows:

- This paper proposes and constructs ARTNET, an interconnected multimodal knowledge infrastructure for the art domain. ARTNET provides a unified organization of graph structure, textual information, and visual content, serving as a foundational carrier for shifting art knowledge representation from isolated structured records toward multimodal semantic modeling.
- This paper establishes a data organization pipeline for multimodal research. Based on nodes and edges exported from a graph database, the pipeline aligns graph chunks, text chunks, and image data at the node level, thereby providing a unified input format for subsequent modeling.
- This paper defines two core tasks for art-oriented scenarios, namely multimodal link prediction and multimodal knowledge graph embedding with edge-type scoring. These tasks are designed to evaluate model capability in capturing edge existence and relation types, respectively.
- This paper further provides baseline methods and evaluation protocols aligned with the above tasks. The link prediction baseline combines multimodal entity representations, GraphSAGE message passing, and an MLP predictor to evaluate the model's ability to distinguish existing edges from non-existing ones. For edge-type modelling, the paper constructs two types of baselines, namely pure KGE and GraphKGE. The former scores triples by combining multimodal entity representations with TransE and ComplEx scoring functions, while the latter further introduces GraphSAGE and R-GCN graph encoders to analyse how general topological structure and relation-aware structural information affect edge-type prediction. Based on unified data splits, global metrics, and art-related subset metrics, the paper systematically evaluates the link prediction, pure KGE, and GraphKGE experiments.

2 Related Work

2.1 Knowledge Graph Datasets

The development of knowledge graph datasets has evolved from general fact organization toward increasingly domain-specific, heterogeneous, and multimodal forms [1]. Early studies mainly rely on open-domain knowledge graphs such as Wikidata [2], DBpedia [3], YAGO [4], and Freebase [5], which represent real-world entities, attributes, and relations in a unified triple-based format. These resources typically emphasize entity scale, relation coverage, and structural consistency, and therefore effectively support classical tasks such as knowledge completion, link prediction, entity alignment, and relation reasoning [6]. As a result, they have established one of the most representative data organization paradigms in knowledge graph research [7]. For example, in the art domain, the triple (*Mona Lisa*, *creator*, *Leonardo da Vinci*) represents the relation between an artwork and its creator, while (*Mona Lisa*, *collection*, *Louvre Museum*) captures the relation between the artwork and its holding institution. This form of representation is highly effective for organizing structured facts and also provides a unified input basis for subsequent knowledge graph learning methods [1].

In the art domain, existing knowledge graph resources have already shown the potential to connect artworks, artists, institutions, materials, styles, and historical context in a structured manner [8], yet they still exhibit several clear limitations. First, many resources are constructed mainly around textual semantics and structured metadata, focusing on fields such as artwork titles, creators, production periods, and collection information, while lacking systematic integration of artwork images [9]. Second, existing art knowledge graphs still often suffer from missing, inconsistent, or weakly structured labels for style, subject matter, period, region, medium, and image theme, which limits their ability to support finer-grained comparison and semantic reasoning over art entities [10], [11]. Third, many current resources are closer to static digital collections or retrieval-oriented resources than to standardized experimental datasets designed for machine learning research. As a result, they often lack consistent standards for data splitting, task formulation, and evaluation metrics [11], [12]. In other words, although existing art knowledge graphs provide valuable cultural knowledge resources, they still fall short of supporting systematic multimodal learning research [12].

From this trajectory, the evolution of knowledge graph datasets is no longer driven solely by the pursuit of larger entity scale, but increasingly by the need to integrate graph structure, textual information, visual evidence, and domain labels within a unified framework, so that the resulting resources can support reusable and comparable research tasks [13]. This shift is particularly important in the art domain, because the core semantics of artistic knowledge are inherently distributed across structural relations, language descriptions, and visual content [12]. Accordingly, building a research-oriented knowledge infrastructure with multimodal support has become a key direction for the further development of art knowledge graph research [13].

2.2 Knowledge Graph Tasks and Methods

From the perspective of task design, knowledge graph research has long centered on entity relation modeling, among which link prediction is one of the most fundamental and representative tasks [14], [15]. This task aims to determine, based on existing graph structure and node representations, whether a certain relation should exist between two entities, or to recover missing edges [14], [16]. Link prediction is not only an important form of knowledge completion, but also a core task for evaluating whether a model can effectively use graph-structured context and node semantic information [14], [15]. In traditional settings, link prediction mainly relies on graph structure or symbolic entity representations [14], [17]. With the development of representation learning, graph neural networks have gradually become an important method for this task, as they learn local structural representations of nodes through neighborhood aggregation and further use them to predict edge existence [16], [18]. For art knowledge graphs, link prediction has direct practical value, because relations between artworks and creators, holding institutions, styles, or subjects are often incomplete, requiring models to recover latent associations from sparse graph structures.

Closely related to edge existence prediction, yet distinct from it, is the task of relation type modeling [15]. This task focuses more on whether a model can correctly identify the semantic relation between a given pair of entities, or rank and score candidate relations [15]. Knowledge graph embedding methods are widely representative in this setting, with typical examples including TransE [19], [25] and ComplEx [20], [25]. The former models the geometric constraint among head entity, relation, and tail entity through a translation assumption. Its formulation is simple and well suited to capturing relatively direct relation patterns [19]. The latter enhances the modeling of complex relations, including symmetry and antisymmetry, through representations in complex space, and therefore shows greater flexibility in multi-relation modeling [20]. Whether the task is entity completion, relation ranking, or triple classification, its essential purpose is to evaluate the ability of a knowledge graph to represent and infer relation semantics [14], [15]. In the art domain, such tasks concern not only whether an edge exists, but also what type of edge it is. They are therefore especially important for understanding diverse semantic relations among artworks, including creation relations, style attribution, material use, and historical association.

In recent years, methods for knowledge graph tasks have shown a growing multimodal trend [11], [13]. Traditional models often treat entities as discrete symbols and rely mainly on entity identifiers and graph structure for modeling, while making limited use of textual descriptions and visual content carried by the entities themselves [21], [22]. With the development of pretrained language models and visual encoders, researchers increasingly incorporate textual semantics, image features, and graph structure into entity representation learning [22], [24]. This enables models not only to understand who an entity is connected to, but also how it is described and what it presents visually [23], [24]. This shift is particularly important for the art domain, because the knowledge carried by artistic objects is not fully contained in symbolic relations alone, but requires image content and descriptive text to jointly supplement its semantic boundaries. Accordingly, multimodal link prediction, multimodal knowledge graph embedding, and cross-modal relation reasoning are gradually becoming more promising directions in art knowledge graph research [13], [23].

2.3 Knowledge Graph Research in the Art Domain

Compared with general knowledge graphs, research on knowledge graphs in the art domain has a stronger demand for multimodal support [26], [27]. The knowledge value of artworks is reflected not only in structured attributes such as creator, period, collection, and material, but also in the objects, composition, color, style, symbolic elements, and historical context contained in the images themselves [27]. Therefore, if an art knowledge graph records only external structural relations, it can usually support the organization and linking of artworks, but cannot adequately support a deeper understanding of their content and cultural meaning [2]. Although existing studies in cultural heritage, art history, and digital humanities have attempted to introduce richer labels, relations, and visual information to improve the semantic expressiveness of art knowledge graphs [31], clear limitations still remain overall. In particular, data construction is often oriented more toward resource organization than experimental research, methods often treat the graph as an auxiliary resource rather than a central research object, and evaluation still lacks unified task definitions and metric systems [13], [29]. Some existing works have begun to combine visual information with semantic relations for object understanding, style analysis, and semantic association modeling, which suggests that image evidence and graph structure are complementary [22], [28]. However, most of these studies still remain at the level of single tasks or local methods [30]. This suggests that the key issue in current art knowledge graph research is no longer simply whether to introduce the image modality, but how to organize multimodal data within a unified framework and make it genuinely support link prediction, relation modeling, and subsequent multimodal learning research [29].

2.4 Difference and Positioning of This Work

Compared with the studies discussed above, the distinctiveness of this work does not lie in proposing only a new link prediction model or a new knowledge graph embedding method, but in reorganizing the problem of art knowledge graphs from the perspective of research infrastructure [26]. The ARTNET proposed in this paper is neither a local extension of existing art graphs nor a simple addition of art images to existing triples. Instead, it brings graph structure, textual information, image information, and label systems into a unified data organization pipeline, so that they can jointly support multimodal link prediction and multimodal KGE edge-type scoring tasks [25], [29]. In other words, the core concern of this work is not a locally optimal result on a single task, but the construction of a scalable, comparable, and reusable multimodal research framework for the art domain.

More specifically, the differences between this work and existing studies are reflected in three aspects. At the data level, this paper further enriches large-scale art graph structure with node-level textual semantics and aligns image information by node, thereby forming multimodal entity representations that can be directly used for modeling. At the task level, this paper no longer treats art knowledge graphs as resources limited to static display or retrieval, but defines two core tasks, namely multimodal link prediction and multimodal edge-type scoring, so that art knowledge graphs become research objects that can be evaluated quantitatively. At the evaluation level, this paper establishes relatively unified experimental settings, data split strategies, and evaluation metrics to support model comparison under a more consistent benchmark. Therefore, the significance of ARTNET lies not only in providing more data, but also in advancing art knowledge graph research from resource organization to multimodal modeling and standardized evaluation. Overall, although existing studies have provided an important foundation for knowledge organization, multimodal modeling, and relation learning in the art domain, they still show clear limitations in the unified organization of multimodal data, the standardized definition of tasks, and the construction of systematic evaluation frameworks [28]. ARTNET is proposed in this context not to replace existing art knowledge resources, but to provide a research infrastructure that supports object understanding, relation reasoning, and cross-modal learning.

3 Methodology

As shown in Figure 1, our method forms an end-to-end framework for constructing and evaluating a multimodal art knowledge graph, with four stages: ARTNET dataset construction, multimodal entity representation, task modelling, and standardized evaluation. In the data construction stage, we collect art-related entities and semantic relations from Wikidata, enrich node information with Wikipedia text and available image resources, and use large-scale graph partitioning to reduce computational and memory costs during full-graph training, enabling subsequent graph algorithms to process large-scale graph data under controlled resources. In the entity representation stage, the model uses text and visual encoders to extract semantic and visual node features, and obtains unified multimodal node representations through projection, missing-modality modelling, and cross-modal fusion. In the task modelling stage, this study focuses on two core tasks. The first is multimodal link prediction based on local subgraph sampling and GraphSAGE message passing to predict connections between node pairs. The second is edge-type modelling for triple-level relation semantics, where pure KGE models score triples with multimodal entity representations and scoring functions such as TransE and ComplEx. GraphKGE further introduces GraphSAGE and R-GCN graph encoders before the KGE scoring layer, allowing entity representations to integrate general topology and relation-aware neighbourhood context, thereby analysing the roles of multimodal content, local graph structure, and relation-type information in edge-type prediction. Finally, during evaluation, we adopt unified data splits, task-specific metrics, and a multi-view evaluation protocol to systematically compare edge existence prediction and edge-type recognition. This framework integrates textual semantics, visual content, and graph structure into a unified modelling pipeline, providing a complete experimental basis for multimodal representation learning and relation modelling in art knowledge graphs.

3.1 Dataset Construction

3.1.1 Construction of the ARTNET Dataset

The ARTNET knowledge graph follows the graph definition $G = \langle N, E\rangle$, where N denotes the set of nodes and E denotes the set of edges. Each node corresponds to an art or knowledge entity and contains structured attributes as well as multimodal information. The dataset construction process consists mainly of two stages: raw graph construction and content enrichment.

First, ARTNET selects art-related nodes from Wikidata as core entities and connects them with associated nodes within a multi-hop relational range, thereby constructing a heterogeneous graph that contains both art nodes and general knowledge nodes. Specifically, the construction starts from art-related categories and iteratively expands along relations to include associated entities as contextual nodes. In subsequent processing, the unique identifier, type labels, and attribute fields of each node are organized as node representations, while the source node, target node, and relation type of each relation are organized as edge representations. This design forms an explicit graph input suitable for subsequent modeling. To control the graph scale, ARTNET only retains relations within three hops. Expansions beyond three hops tend to include a large portion of Wikidata, and existing studies show that such expansions do not improve model accuracy for most tasks [44], [45]. After graph construction, relation types are filtered, and relations with too few instances are removed to reduce instability in subsequent modeling.

On this basis, this paper enriches the textual information of nodes. Since textual descriptions in the raw graph are usually short and cannot fully express the semantic information of art objects, this paper uses external knowledge base identifiers as bridges to map graph entities to Wikipedia pages. For entities that can be matched to valid pages, the introductory paragraphs and main body content are extracted, cleaned, and saved as enriched text. For entities without available pages, the basic descriptions from their original attributes are retained.

In addition, this paper defines only nodes with available images as art nodes and downloads their associated images as visual modality inputs. To meet the input requirements of different visual encoding models, all images are uniformly resized along the longest edge to 64, 256, and 512 pixels, which supports different visual feature extraction methods while retaining a higher-resolution version for future fine-grained visual content modeling. Nodes without images are not used as art nodes in subsequent task partitioning. Instead, they are retained as general knowledge nodes and participate in modeling only as auxiliary context in the graph structure. Although these nodes lack visual information, they still preserve textual attributes and relational connections, which provide semantic supplements and structural support for art nodes. Through the above processing, ARTNET preserves the structure of the original knowledge graph while forming a unified data representation in which art nodes carry image and text information, and knowledge nodes provide structural and semantic assistance. This representation provides the basis for subsequent multimodal representation learning and relation modeling.

3.1.2 Large-Scale Graph Chunking and Cross-Chunk Subgraph Design

Since the complete graph is too large to be loaded and directly processed as a whole by subsequent graph algorithms, this paper partitions the entire graph into multiple chunks according to node order and stores the corresponding node files and edge files separately. After chunking, each node belongs to only one fixed chunk, while each edge is stored according to the chunk of its source node. In this way, the original large-scale graph is divided into a set of smaller subgraphs with explicit boundaries. This design does

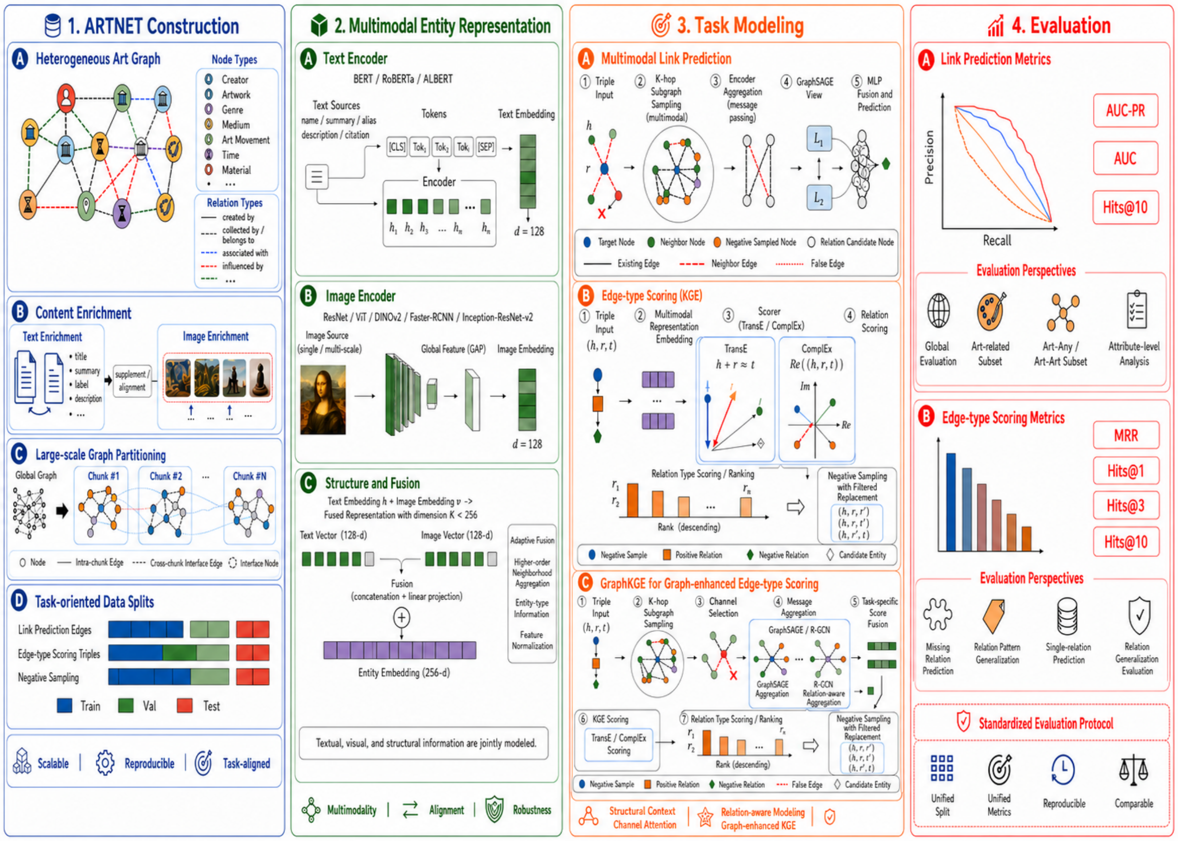

Fig. 1. Overall framework of the proposed ARTNET methodology.

not merely localize the graph. Instead, it converts the large-scale graph into a staged data representation, allowing graph algorithms to be progressively applied to the whole graph at the chunk level. Since connections still exist across different chunks, this paper further preserves one-hop cross-chunk contextual information so that structural dependencies between adjacent chunks are not entirely lost during partitioning. Through this design, the large-scale graph is organized into a chunked structure suitable for staged input to graph algorithms, thereby providing a foundation for subsequent feature extraction, graph structural propagation, and edge modeling.

3.2 Algorithmic Framework

3.2.1 Multimodal Node Feature Representation

The multimodal feature representation of nodes consists of text feature extraction, image feature extraction, and cross-modal feature fusion. Its goal is to learn a unified, robust, and discriminative initial representation for each node in the graph. Let the textual description of node i be x_i^t and its image input be x_i^v. This paper uses a text encoder and a visual encoder to learn representations from the two modalities, respectively.

For the textual modality, this paper adopts pretrained language models to encode the entity descriptions of nodes. In the experiments, BERT, RoBERTa, and ALBERT are used as text backbones to compare the semantic modeling capability of different pretrained language models. Given the textual input x_i^t of node i, its textual representation is defined as

$$\mathbf{h}_i^t = f_t(x_i^t),$$

where $f_t(\cdot)$ denotes the text encoder, and $\mathbf{h}_i^t \in \mathbb{R}^{d_t}$ denotes the textual feature vector of the node. Since entity descriptions usually contain category information, semantic attributes, and contextual knowledge, the textual modality provides a stable semantic prior for nodes. Therefore, this paper retains the text branch as an essential component of node representation.

For the visual modality, this paper uses multiple visual backbones to extract visual features from entity-associated images, so as to enhance the ability of node representations to model appearance information, visual semantics, and local details. Specifically, the visual encoders used in the experiments include ResNet-50, ResNet-18, VGG-19, DINOv2-base, Faster R-CNN ResNet50 FPN V2, and Inception-ResNet-v2. Except for one comparative setting in which ResNet-18 is randomly initialized and trainable, all other visual encoders use pretrained parameters and remain frozen. This setting highlights the effect of different visual representation capabilities on the quality of node representations. For node i with an available image, its visual representation is defined as

$$\mathbf{h}_i^v = f_v(x_i^v),$$

where $f_v(\cdot)$ denotes the visual encoder, and $\mathbf{h}_i^v \in \mathbb{R}^{d_v}$ denotes the visual feature vector of the node.

Since some nodes in the real data lack available images, directly discarding these nodes would lead to the loss of graph structural information and node content information. Replacing missing images with zero vectors may also confuse the absence of a modality with a weak visual response. To address this issue, this paper explicitly models missing images during the fusion stage. Let the binary indicator $m_i \in \{0, 1\}$ denote the image availability of node i, where $m_i = 1$ indicates that the node has a real image and $m_i = 0$ indicates that the image is missing. A learnable missing-image embedding $\mathbf{e}_{extmiss} \in \mathbb{R}^{d'}$ is introduced to represent the special state of having no image information. First, the textual and visual features are mapped into a unified dimensional space:

$$\hat{\mathbf{h}}_i^t = \mathbf{W}_t \mathbf{h}_i^t + \mathbf{b}_t, \qquad \hat{\mathbf{h}}_i^v = \mathbf{W}_v \mathbf{h}_i^v + \mathbf{b}_v,$$

where $\mathbf{W}_t$ and $\mathbf{W}_v$ denote the linear projection matrices for the textual and visual modalities, respectively. The effective visual representation of the node is then defined as

$$\tilde{\mathbf{h}}_i^v = m_i \hat{\mathbf{h}}_i^v + (1 - m_i)\mathbf{e}_{\text{miss}}.$$

This design allows nodes with real images to use actual visual features in fusion, while nodes with missing images are complemented by a learnable embedding. As a result, all nodes can participate in representation learning, and invalid visual noise is prevented from interfering with the fusion result.

In the cross-modal fusion stage, this paper feeds the projected textual feature, the effective visual feature, and the image availability mask into the fusion module. This enables the model to capture the complementary relationship between text and images while explicitly perceiving whether a node has real visual evidence. The fusion input is defined as

$$\mathbf{u}_i = [\hat{\mathbf{h}}_i^t \,\|\, \tilde{\mathbf{h}}_i^v \,\|\, m_i],$$

where $\|$ denotes vector concatenation. A multilayer perceptron is then used to perform nonlinear transformation on the concatenated multimodal features, producing the unified node representation:

$$\mathbf{z}_i = \mathbf{W}_2 \, \mathrm{Dropout}\left(\mathrm{ReLU}(\mathbf{W}_1 \mathbf{u}_i + \mathbf{b}_1)\right) + \mathbf{b}_2,$$

where $\mathbf{z}_i$ denotes the final multimodal representation of the node. Compared with simple concatenation or linear weighting, the MLP-based fusion mechanism can better learn nonlinear interactions between cross-modal features, thereby strengthening the complementary modeling of textual semantics and visual content. By introducing an explicit missingness mask and a learnable missing-image embedding, the model can distinguish between nodes without images and nodes with images but weak visual information, which improves the robustness and discriminative ability of node representations under incomplete modality conditions.

3.2.2 Graph-Based Link Prediction Algorithm

This paper adopts a GraphSAGE link prediction framework based on local subgraph modeling to determine whether an edge exists between a given pair of nodes. For real edges in the training graph, the model treats them as positive samples and constructs negative samples by replacing one endpoint of an edge. To ensure the validity of negative samples, each generated candidate edge must not form a self-loop and must not overlap with any existing edge in the global set of real edges. In this way, the link prediction task is formulated as a binary classification problem over positive and negative edge samples.

At each training step, the model first takes the union of the endpoints of the positive and negative edges in the current batch as the seed node set. It then performs multi-hop neighborhood sampling around these seed nodes only on the propagation graph constructed from training positive edges, thereby building a local subgraph related to the current candidate edges. Unlike end-to-end propagation over the complete graph, this strategy restricts computation to the local structural region around the target edges. It enables the model to control the computational cost of large-scale graph learning while using the topological context that is most relevant to the current prediction task.

To avoid label leakage, the supervised positive edges in the current batch are explicitly removed from the local message graph before graph neural network propagation. As a result, the model cannot directly access the target edges during the forward pass and must infer edge existence from node attributes and surrounding neighborhood structures. After local graph propagation, the model obtains structure-enhanced representations for the two endpoints of each candidate edge. These two representations are concatenated and fed into an edge scorer to produce the prediction score for edge existence. Given a candidate edge $e = (u, v)$, its score is defined as

$$s(u, v) = \mathrm{MLP}([\mathbf{h}_u \| \mathbf{h}_v]),$$

where $[\cdot\|\cdot]$ denotes vector concatenation, and $\mathbf{h}_u$ and $\mathbf{h}_v$ denote the final representations of nodes u and v after graph propagation, respectively.

During training, real edges are labeled as positive samples, and constructed edges are labeled as negative samples. The model is optimized with a binary classification loss:

$$\mathcal{L} = -\sum_{(u,v)\in B^+} \log \sigma(s(u, v)) - \sum_{(u,v)\in B^-} \log(1 - \sigma(s(u, v))),$$

where B^+ and B^- denote the sets of positive and negative edges in the current batch, respectively, and $\sigma(\cdot)$ denotes the sigmoid function.

Overall, the graph-based link prediction algorithm in this paper consists of positive and negative sample construction, local subgraph extraction, supervised edge masking, graph structural propagation, and edge score prediction. Built on multimodal node representations, this method integrates node-level semantics with local graph structural information, thereby enabling effective modeling of potential connections between nodes.

3.2.3 Graph-Based Feature Propagation

After multimodal node feature fusion, this paper further performs graph structural feature propagation on local subgraphs to enhance the ability of node representations to capture neighborhood topological context. Let the local subgraph corresponding to the current batch be $G_s = (V_s, E_s')$, where V_s denotes the node set composed of seed nodes and their sampled neighbors, and E_s' denotes the message-passing edge set after removing the current supervised positive edges. For any node $i \in V_s$, its multimodal fused representation is denoted as $\mathbf{z}_i$, which serves as the input to the graph neural network:

$$\mathbf{h}_i^{(0)} = \mathbf{z}_i.$$

This paper adopts GraphSAGE as the graph encoder. It aggregates neighborhood features layer by layer on the local subgraph and updates each node representation by combining the aggregated neighborhood information with the node's own representation. Let $\mathcal{N}(i)$ denote the neighbor set of node i at layer l. The neighborhood aggregation result is defined as

$$\mathbf{m}_i^{(l)} = \mathrm{Mean}(\{\mathbf{h}_j^{(l)} \mid j \in \mathcal{N}(i)\}).$$

The node representation is then updated as

$$\mathbf{h}_i^{(l+1)} = \phi(\mathbf{W}_{\text{self}}^{(l)}\mathbf{h}_i^{(l)} + \mathbf{W}_{\text{nbr}}^{(l)}\mathbf{m}_i^{(l)} + \mathbf{b}^{(l)}),$$

where $\mathbf{W}_{\text{self}}^{(l)}$ and $\mathbf{W}_{\text{nbr}}^{(l)}$ denote learnable parameters applied to the node's own features and the aggregated neighborhood features, respectively, $\mathbf{b}^{(l)}$ denotes the bias term, and $\phi(\cdot)$ denotes the nonlinear activation function. Through this update, the node representation progressively absorbs neighborhood structural information at each layer and integrates it with the original multimodal semantic information.

After multi-layer message passing, each node obtains a structure-enhanced representation. This representation preserves the textual and visual semantics of the node itself and further incorporates relation patterns and topological context from the local neighborhood. It is therefore no longer a purely attribute-based representation, but a unified representation that encodes both content information and structural information. Since this feature propagation is performed only on local subgraphs dynamically constructed around candidate edges, the structural information absorbed by the model is more closely related to the current prediction target, thereby providing more discriminative input features for subsequent edge existence prediction.

3.2.4 KGE-Based Edge-Type Modeling

Pure KGE. This paper further conducts edge-type modeling experiments and evaluates them as an independent knowledge graph embedding task. Unlike link prediction, which mainly focuses on whether a connection exists between a pair of nodes, this task further considers the relation type of each edge. A typed edge is represented as a triple (h, r, t), and knowledge graph embedding is performed based on multimodal entity representations and relation embeddings. The model does not perform neighbor sampling or introduce message passing with graph neural networks. Instead, it directly assigns scores to triples.

At the entity representation level, this paper still uses the preceding multimodal encoding module to construct a content representation for each entity. Let the fused multimodal representation of entity i be $\mathbf{z}_i$. Before entering the KGE layer, the model first projects it into the knowledge graph embedding space through a linear mapping:

$$\mathbf{e}_i = \mathbf{W}_k\mathbf{z}_i + \mathbf{b}_k.$$

On this basis, the model learns a relation embedding $\mathbf{r}$ for each relation $r \in \mathcal{R}$. Therefore, the modeling target is no longer a binary classification problem over node pairs, but a triple scoring problem jointly defined by the head entity representation $\mathbf{e}_h$, the relation representation $\mathbf{r}$, and the tail entity representation $\mathbf{e}_t$.

For relation scoring, this paper selects TransE and ComplEx as two representative KGE methods for comparison. For TransE, the model assumes that the head entity should be close to the tail entity after translation by the relation. Its scoring function is defined as

$$s(h, r, t) = -\|\mathbf{e}_h + \mathbf{r} - \mathbf{e}_t\|_2.$$

This formulation is simple and provides a stable basic baseline for relation modeling. For ComplEx, the model represents entities and relations in the complex space. Let the entity and relation representations be decomposed into real and imaginary parts. Its scoring function is written as

$$\begin{aligned} s(h, r, t) = \sum_k \big(& e_{h,k}^{\mathrm{Re}} r_k^{\mathrm{Re}} e_{t,k}^{\mathrm{Re}} + e_{h,k}^{\mathrm{Im}} r_k^{\mathrm{Re}} e_{t,k}^{\mathrm{Im}} \\ &+ e_{h,k}^{\mathrm{Re}} r_k^{\mathrm{Im}} e_{t,k}^{\mathrm{Im}} - e_{h,k}^{\mathrm{Im}} r_k^{\mathrm{Im}} e_{t,k}^{\mathrm{Re}}\big). \end{aligned}$$

Compared with TransE, ComplEx can more naturally represent asymmetric relations and more complex relation patterns, making it more suitable for analyzing the effect of relation semantic modeling capability on the results.

During training, the model constructs positive samples from real triples and generates negative samples by replacing the head entity or the tail entity. In this way, the task is formulated as a discrimination problem between real triples and corrupted triples. Let the positive sample score be s^+ and the negative sample score be s^-. The training objective is defined as

$$\mathcal{L} = -\frac{1}{N_+}\sum_{i=1}^{N_+}\log\sigma(s_i^+) - \frac{1}{N_-}\sum_{j=1}^{N_-}\log(1 - \sigma(s_j^-)),$$

where $\sigma(\cdot)$ denotes the sigmoid function. The objective increases the scores of real triples and suppresses the scores of corrupted triples, thereby learning stable entity and relation representations.

During inference, the model can directly output a plausibility score for a given triple (h, r, t). It can also score and rank all candidate relations under a fixed entity pair (h, t) to perform edge-type prediction.

GraphKGE. Based on the pure KGE experiments, this paper further constructs GraphKGE to examine the effect of local graph structural information on edge-type modeling. Unlike pure KGE, which directly scores triples based on multimodal content representations of entities, GraphKGE introduces a graph neural network encoder before the KGE scoring layer. This design allows entity representations to incorporate contextual information from neighboring nodes and local connection structures. The method still uses typed triples (h, r, t) as supervision targets, and its scoring functions and training objective remain consistent with those of pure KGE. The main difference lies in the construction of entity representations.

Let $\mathbf{z}_i$ denote the content representation of entity i obtained after multimodal encoding and fusion. GraphKGE first samples a K-hop neighborhood subgraph around the head and tail entities involved in the current batch:

$$\mathcal{G}_s = (\mathcal{V}_s, \mathcal{E}_s),$$

where $\mathcal{V}_s$ contains the head and tail entities in the batch as well as their neighboring nodes, and $\mathcal{E}_s$ denotes the edges in the sampled subgraph. The initial node feature is given by the multimodal fused representation:

$$\mathbf{h}_i^{(0)} = \mathbf{z}_i, \quad i \in \mathcal{V}_s.$$

The model then performs message passing on the subgraph:

$$\mathbf{h}_i^{(l+1)} = \mathrm{GNN}^{(l)}\big(\mathbf{h}_i^{(l)}, \{\mathbf{h}_j^{(l)} : j \in \mathcal{N}(i)\}\big),$$

where $\mathcal{N}(i)$ denotes the neighbor set of node i. After several propagation layers, the model obtains the structure-enhanced representation $\mathbf{g}_i$ and maps it into the KGE embedding space:

$$\mathbf{e}_i = \mathbf{W}_k\mathbf{g}_i + \mathbf{b}_k.$$

TABLE 1
Core Statistics of the ARTNET Dataset

Nodes	Edges	Unique Node Pairs	Text Nodes	Image Nodes	ART Nodes
1,706,516	14,382,036	12,057,700	1,702,213	448,349	431,773

Therefore, the entity embedding in GraphKGE contains both the entity's own multimodal semantics and local graph structural information.

This paper uses GraphSAGE and R-GCN as graph encoders. GraphSAGE does not explicitly distinguish relation types. Instead, it aggregates representations from neighboring nodes to examine the effect of general topological structure:

$$\mathbf{h}_i^{(l+1)} = \sigma\big(\mathbf{W}^{(l)} \cdot \mathrm{AGG}(\mathbf{h}_i^{(l)}, \{\mathbf{h}_j^{(l)} : j \in \mathcal{N}(i)\})\big),$$

where $\mathrm{AGG}(\cdot)$ denotes the neighborhood aggregation function, and $\sigma(\cdot)$ denotes the nonlinear activation function.

R-GCN further introduces relation-type information into message passing. Its update process is defined as

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{r\in\mathcal{R}}\sum_{j\in\mathcal{N}_i^r}\frac{1}{c_{i,r}}\mathbf{W}_r^{(l)}\mathbf{h}_j^{(l)} + \mathbf{W}_0^{(l)}\mathbf{h}_i^{(l)}\right),$$

where $\mathcal{N}_i^r$ denotes the set of neighbors connected to node i through relation r, $\mathbf{W}_r^{(l)}$ denotes the relation-specific propagation parameter, and $c_{i,r}$ denotes the normalization coefficient. Compared with GraphSAGE, R-GCN can encode relation-aware structural information during entity representation learning.

To avoid label leakage, this paper masks supervised edges in the training and evaluation subgraphs. When a batch contains a target entity pair (h, t) to be scored, the model removes the corresponding query edge and its reverse edge from the message-passing graph. As a result, the graph encoder can only rely on the remaining neighborhood structure and entity semantics for inference.

Finally, GraphKGE follows the TransE or ComplEx scoring functions and the positive-negative sample discrimination objective used in pure KGE to score and optimize triples (h, r, t). Therefore, GraphKGE does not change the KGE task itself, but enhances entity representations through graph neural networks. By comparing pure KGE, GraphSAGE-KGE, and R-GCN-KGE, this paper analyzes the different contributions of multimodal entity content, general topological structure, and relation-aware structural information to edge-type prediction.

3.3 Experimental Setup

3.3.1 Dataset Description

ARTNET is a multimodal knowledge graph dataset constructed in this paper for the art domain, supporting tasks such as link prediction and edge-type modeling. The dataset centers on art-related entities and their semantic relations, and organizes graph structural information, textual descriptions, and visual information within a unified framework. It therefore provides a basis for art object understanding, relation modeling, and multimodal representation learning. Unlike conventional knowledge graphs that contain only structured relations, ARTNET preserves the entity-relation structure while incorporating textual and visual modality information at the node level. This allows the dataset to represent the semantic content and contextual associations of art entities more comprehensively. The core statistics of the dataset are shown in Table 1.

3.3.2 Evaluation Metrics

This paper defines evaluation metrics separately for the link prediction task and the edge-type modeling task. For link prediction, this paper uses AUC, AUPRC, and Hit@10 for evaluation. AUC measures the overall ability of the model to distinguish positive edges from negative edges. AUPRC focuses more on the identification quality of real edges, while Hit@10 characterizes the ranking performance of real edges among candidate results. Since AUPRC is more sensitive to positive sample identification, this paper uses it as the primary model selection metric in the link prediction experiments.

For edge-type modeling, this paper evaluates model performance from two perspectives: relation ranking and triple discrimination. For relation ranking, MRR, Hits@1, Hits@3, Hits@10, and Top-1 Accuracy are used to measure the ranking position of the true relation in the candidate relation set and whether it is ranked first. For triple discrimination, AUC and AUPRC are used to evaluate the ability of the model to distinguish real triples from negative triples. Through these metrics, this paper provides a comprehensive evaluation of model performance from the perspectives of edge existence prediction and relation type identification.

3.3.3 Comparison Experiments

This paper sets up two groups of comparison experiments on the same dataset: link prediction and edge-type scoring. For link prediction, eight multimodal combinations are used to compare the effects of different text backbones, image backbones, and weight strategies on edge existence prediction. Based on the link prediction results, this paper selects the four best-performing multimodal combinations as the basis for entity representation and further conducts edge-type scoring experiments with two scorers, TransE and ComplEx. Therefore, the edge-type scoring experiments contain eight settings in total. The experimental combinations and their corresponding tasks are shown in Table 2.

3.3.4 Experimental Parameter Settings

The link prediction experiments use deduplicated undirected edge data, which is split into training, validation, and test sets at a ratio of 8:1:1. During training, the model uses two-hop local subgraph sampling with the number of sampled neighbors set to $[8, 3]$. Each step samples 256 positive edges and constructs negative edges at a ratio of 1:1, with the maximum number of training steps set to 10000. Graph propagation is performed only on positive edges from the training set, and the supervised edges in the current batch are removed before propagation. The initial node representation is obtained by projecting text features and image features into 128-dimensional spaces and then fusing them into a 256-dimensional representation. The graph encoder uses a two-layer GraphSAGE with a hidden dimension of 256 and a

TABLE 2
Comparison Settings for Link Prediction and Edge-Type Scoring Experiments

Model	Weight Strategy	Link Prediction	TransE	ComplEx
BERT + ResNet-50 [46]	Pretrained, Frozen	✓	✓	✓
BERT + ResNet-18	Pretrained, Frozen	✓		
BERT + ResNet-18	Random Init, Trainable	✓		
BERT + VGG-19 [47]	Pretrained, Frozen	✓		
BERT + DINOv2-base	Pretrained, Frozen	✓	✓	✓
RoBERTa + ResNet-50 [48]	Pretrained, Frozen	✓	✓	✓
BERT + Faster R-CNN ResNet50 FPN V2 [49]	Pretrained, Frozen	✓	✓	✓
ALBERT + Inception-ResNet-v2 [50]	Pretrained, Frozen	✓		

dropout rate of 0.1. The edge predictor is a two-layer MLP, and the loss function is `BCEWithLogitsLoss`. Training adopts an early stopping strategy with patience set to 10, and `val_aucpr` is used as the model selection metric.

3.3.5 Implementation Environment

The experiments use Python as the main development language and manage model parameters, data paths, training settings, and evaluation settings through JSON configuration files. The data loading pipeline uses a Neo4j graph database, and the Python implementation retrieves node, edge, and relation-type information by executing Cypher queries through the Neo4j Driver. Text encoders are downloaded and loaded through Hugging Face Transformers, including BERT, RoBERTa, and ALBERT, with Transformers version 4.40.2. Image encoders are loaded and used through Torchvision, including ResNet and VGG, with Torchvision version 0.16.0. Deep learning model training is implemented with PyTorch 2.1.0+cu121, and the graph neural network modules are implemented with PyTorch Geometric 2.5.3. The experiments run on Ubuntu 22.04.5 LTS with Python 3.11.14. The machine has 96 CPU cores, 503 GB of memory, and an NVIDIA GeForce RTX 4080 SUPER GPU with 31.47 GB of video memory. The CUDA driver version is 13.0, the CUDA version used by PyTorch is 12.1, and the cuDNN version is 8902. Other main dependencies include NumPy 1.26.4, Scikit-learn 1.8.0, Matplotlib 3.10.8, Pillow 12.1.1, Neo4j Python Driver 6.1.0, Requests 2.32.5, BeautifulSoup 4.14.3, lxml 6.0.2, and tqdm 4.67.3.

The KGE experiments use a relation-stratified 8:1:1 split for triples. The training parameters are set uniformly across runs. The key settings are `batch_size`=512, `num_neg`=1, `epochs`=3, `eval_batch_size`=128, and `max_steps`=0. During validation, at most 500000 triples are used, and `val_mrr` serves as the model selection metric. For each positive triple, the model constructs one negative sample by replacing either the head entity or the tail entity with probability 0.5. The loss function is `BCEWithLogitsLoss`. All KGE experiments exclude graph structural propagation. Entity embeddings are constructed only from text features, image features, and their fused representations. The results are then compared under the TransE and ComplEx scorers.

4 Results

This section reports the experimental results on the ARTNET dataset, covering three main groups of experiments: multimodal link prediction, multimodal KGE edge-type scoring, and GraphKGE edge-type scoring with graph encoders. The results are presented using both full test-set metrics and Art-Any subset metrics. Art-Any denotes the subset in which at least one node in the candidate node pair is an art-related node with both textual and visual information.

4.1 Multimodal Link Prediction Results

Table 3 reports the GraphSAGE-based multimodal link prediction results on the ARTNET dataset and compares different combinations of text and visual encoders.

On the full test set, RoBERTa + ResNet-50 achieves the best results across AUC-PR, AUC, and Hit@10, with scores of 0.1457, 0.9227, and 0.7606, respectively. ALBERT + Inception-ResNet-v2 ranks second in AUC-PR with a score of 0.1380, while obtaining an AUC of 0.9064 and a Hit@10 of 0.6562. BERT + DINOv2, BERT + ResNet-50, and BERT + ResNet-18 (pretrained) achieve full-set AUC-PR scores of 0.1159, 0.1131, and 0.1131, respectively, placing them in the middle range. BERT + ResNet-18 (trainable) obtains the lowest full-set AUC-PR score, with a value of 0.0948.

On the Art-Any subset, RoBERTa + ResNet-50 also achieves the best performance, with an Art-Any AUC-PR of 0.4186 and an Art-Any Hit@10 of 0.8992. BERT + Faster R-CNN ResNet50 FPN V2 ranks second in Art-Any AUC-PR with a score of 0.4019, and its Art-Any Hit@10 reaches 0.8638. BERT + DINOv2 obtains an Art-Any AUC-PR of 0.3766 and an Art-Any Hit@10 of 0.7987. BERT + ResNet-18 (trainable) gives the lowest Art-Any AUC-PR score, with a value of 0.3229.

Compared with the full-set metrics, all models achieve higher AUC-PR and Hit@10 scores on the Art-Any subset. For example, the AUC-PR of RoBERTa + ResNet-50 increases from 0.1457 to 0.4186, and its Hit@10 increases from 0.7606 to 0.8992. For BERT + Faster R-CNN ResNet50 FPN V2, AUC-PR increases from 0.1093 to 0.4019, and Hit@10 increases from 0.6409 to 0.8638.

4.2 Multimodal KGE Edge-Type Scoring Results

Table 4 reports the results of the multimodal KGE edge-type scoring experiment. This experiment represents relation-typed edges as triples and compares the performance of TransE and ComplEx scoring functions under different multimodal entity representations.

TABLE 3
Link prediction results on the ARTNET dataset

Model	AUC-PR	AUC	Hit@10	Art-Any AUC-PR	Art-Any Hit@10
RoBERTa + ResNet-50	0.1457	0.9227	0.7606	0.4186	0.8992
ALBERT + Inception-ResNet-v2	0.1380	0.9064	0.6562	0.3675	0.7600
BERT + DINOv2	0.1159	0.9018	0.6457	0.3766	0.7987
BERT + ResNet-50 (pretrained)	0.1131	0.9020	0.6592	0.3698	0.8154
BERT + Faster R-CNN ResNet50 FPN V2	0.1093	0.8977	0.6409	0.4019	0.8638
BERT + VGG-19	0.1039	0.8939	0.6318	0.3521	0.8175
BERT + ResNet-18 (pretrained)	0.1036	0.8991	0.6428	0.3299	0.8173
BERT + ResNet-18 (trainable)	0.0948	0.8934	0.6357	0.3229	0.8124

TABLE 4
Comparison of multimodal KGE edge-type scoring results on the ARTNET dataset.

Scorer	Model	MRR	Hit@10	AUC-PR	Art-Any MRR	Art-Any AUC-PR
ComplEx	ALBERT + Inception-ResNet-v2	0.5104	0.8259	0.9601	0.7046	0.8747
ComplEx	BERT + DINOv2-base	0.4796	0.7940	0.9708	0.6316	0.9014
ComplEx	RoBERTa + ResNet-50	0.4631	0.7685	0.9868	0.6613	0.9548
ComplEx	BERT + ResNet-50	0.4304	0.7638	0.9686	0.6046	0.9118
TransE	RoBERTa + ResNet-50	0.3680	0.5809	0.8958	0.6320	0.6920
TransE	ALBERT + Inception-ResNet-v2	0.3413	0.5633	0.8737	0.5939	0.6671
TransE	BERT + ResNet-50	0.3310	0.5253	0.8826	0.6248	0.6774
TransE	BERT + DINOv2-base	0.2605	0.4412	0.8569	0.4554	0.6780

Under the ComplEx scorer, ALBERT + Inception-ResNet-v2 achieves the highest MRR and Hit@10, with values of 0.5104 and 0.8259, respectively. RoBERTa + ResNet-50 obtains the highest AUC-PR and Art-Any AUC-PR, reaching 0.9868 and 0.9548, respectively. BERT + DINOv2-base also shows strong discriminative performance, with an AUC-PR of 0.9708, an Art-Any AUC-PR of 0.9014, an MRR of 0.4796, and a Hit@10 of 0.7940. BERT + ResNet-50 obtains an MRR of 0.4304 and a Hit@10 of 0.7638 within the ComplEx group.

Under the TransE scorer, RoBERTa + ResNet-50 achieves the best results in this group on full-set MRR, Hit@10, and AUC-PR, with values of 0.3680, 0.5809, and 0.8958, respectively. ALBERT + Inception-ResNet-v2 obtains an MRR of 0.3413 and a Hit@10 of 0.5633. BERT + ResNet-50 reaches an MRR of 0.3310 and a Hit@10 of 0.5253. BERT + DINOv2-base gives the lowest MRR and Hit@10 within the TransE group, with values of 0.2605 and 0.4412, respectively.

The comparison between scoring functions shows that ComplEx consistently outperforms TransE on MRR, Hit@10, and AUC-PR. For example, ComplEx with ALBERT + Inception-ResNet-v2 obtains an MRR of 0.5104, whereas TransE with the same multimodal representation reaches 0.3413. Similarly, ComplEx with RoBERTa + ResNet-50 achieves an AUC-PR of 0.9868, while the corresponding TransE variant obtains 0.8958. These results suggest that the bilinear complex-space formulation of ComplEx is more effective than the translational assumption of TransE for modeling relation types in ARTNET.

On the Art-Any subset, most model combinations achieve a higher Art-Any MRR than their full-set MRR. For instance, the MRR of ComplEx with ALBERT + Inception-ResNet-v2 increases from 0.5104 to 0.7046, and the MRR of TransE with RoBERTa + ResNet-50 increases from 0.3680 to 0.6320. In terms of Art-Any AUC-PR, ComplEx with RoBERTa + ResNet-50 achieves the highest value of 0.9548, followed by ComplEx with BERT + ResNet-50 at 0.9118 and ComplEx with BERT + DINOv2-base at 0.9014. This pattern indicates that multimodal KGE models, especially those based on ComplEx, remain effective when evaluation focuses on art-related relation prediction.

4.3 GraphKGE Edge-Type Scoring Results

Table 5 reports the edge-type scoring results under the GraphKGE setting. This experiment further introduces a GraphSAGE or R-GCN graph encoder on top of multimodal entity representations, and combines it with either TransE or ComplEx scoring functions for relation-type prediction.

Under the ComplEx scorer, ALBERT + Inception-ResNet-v2 + R-GCN frozen achieves the highest MRR and Hit@10, with values of 0.4785 and 0.8232, respectively. RoBERTa + ResNet-50 + GraphSAGE frozen obtains the highest AUC-PR and Art-Any AUC-PR, reaching 0.9853 and 0.9428, respectively. RoBERTa + ResNet-50 + R-GCN frozen also performs strongly, with an MRR of 0.4666, an AUC-PR of 0.9813, and an Art-Any MRR of 0.6909.

Under the TransE scorer, ALBERT + Inception-ResNet-v2 + R-GCN frozen achieves the highest MRR and Art-Any MRR, with values of 0.5264 and 0.8136, respectively. RoBERTa + ResNet-50 + R-GCN frozen obtains an MRR of 0.5001 and an Art-Any MRR

TABLE 5
Comparison of GraphKGE edge-type scoring results on the ARTNET dataset

Scorer	Model	MRR	Hit@10	AUC-PR	Art-Any MRR	Art-Any AUC-PR
ComplEx	ALBERT + Inception-ResNet-v2 + R-GCN frozen	0.4785	0.8232	0.9629	0.5453	0.8906
ComplEx	RoBERTa + ResNet-50 + R-GCN frozen	0.4666	0.7539	0.9813	0.6909	0.9385
ComplEx	ALBERT + Inception-ResNet-v2 + GraphSAGE frozen	0.4120	0.7902	0.9718	0.4755	0.9119
ComplEx	RoBERTa + ResNet-50 + GraphSAGE frozen	0.3653	0.6744	0.9853	0.4752	0.9428
TransE	ALBERT + Inception-ResNet-v2 + R-GCN frozen	0.5264	0.6591	0.8816	0.8136	0.6741
TransE	RoBERTa + ResNet-50 + R-GCN frozen	0.5001	0.6395	0.8861	0.8037	0.6842
TransE	RoBERTa + ResNet-50 + GraphSAGE frozen	0.3217	0.5983	0.8817	0.5084	0.6591
TransE	ALBERT + Inception-ResNet-v2 + GraphSAGE frozen	0.3192	0.5397	0.8879	0.5591	0.6891

of 0.8037. In contrast, the two GraphSAGE frozen variants obtain lower MRR values of 0.3217 and 0.3192.

The comparison between graph encoders shows that R-GCN generally outperforms GraphSAGE in terms of MRR and Art-Any MRR under the same scoring function and comparable multimodal backbones. For example, under the TransE + ALBERT + Inception-ResNet-v2 setting, R-GCN obtains an MRR of 0.5264, whereas GraphSAGE reaches 0.3192. Under the TransE + RoBERTa + ResNet-50 setting, R-GCN obtains an MRR of 0.5001, while GraphSAGE reaches 0.3217. These results suggest that relation-aware graph encoding provides stronger ranking performance for edge-type prediction in ARTNET.

On the Art-Any subset, all GraphKGE models achieve higher Art-Any MRR than their corresponding full-set MRR. TransE + ALBERT + Inception-ResNet-v2 + R-GCN frozen obtains the highest Art-Any MRR of 0.8136, followed by TransE + RoBERTa + ResNet-50 + R-GCN frozen with 0.8037. In terms of Art-Any AUC-PR, ComplEx + RoBERTa + ResNet-50 + GraphSAGE frozen achieves the highest value of 0.9428. This pattern indicates that GraphKGE models remain particularly effective when the evaluation focuses on art-related relation prediction, while different metrics favor different graph encoders and scoring functions.

Comparing pure KGE with GraphKGE shows that introducing a graph encoder does not provide a consistent advantage across all metrics, but it leads to a clearer improvement in relation ranking, especially on the Art-Any MRR metric. The best Art-Any MRR in pure KGE is 0.7046, whereas TransE + ALBERT + Inception-ResNet-v2 + R-GCN frozen achieves an Art-Any MRR of 0.8136 in the GraphKGE setting, and TransE + RoBERTa + ResNet-50 + R-GCN frozen also reaches 0.8037. These results indicate that relation-aware graph structural encoding can effectively improve the model's ability to rank art-related relations.

At the same time, pure KGE remains highly competitive on discriminative metrics such as AUC-PR and Art-Any AUC-PR. For example, in pure KGE, ComplEx + RoBERTa + ResNet-50 achieves an AUC-PR of 0.9868 and an Art-Any AUC-PR of 0.9548, both of which are higher than the best results obtained by GraphKGE. Therefore, the main value of GraphKGE is not to fully replace pure KGE, but to improve performance in art-related edge-type ranking scenarios by introducing structured neighborhood information. This finding further suggests that relation-aware graph encoders such as R-GCN are better suited to capturing the complex relational context of ARTNET.

5 Evaluation

5.1 Strengths

This section discusses the experimental results from three perspectives: evaluation metrics, test subsets, and model performance. The paper adopts task-specific evaluation protocols. For link prediction, AUC, AUC-PR, and Hits@K are used to measure the ability to distinguish real edges from negative edges, the quality of positive-edge identification, and the ranking performance of candidate edges. For edge-type prediction, MRR, Hits@10, AUC, and AUC-PR are used to evaluate relation-type modeling from the perspectives of relation ranking and triple classification. The paper also reports both full-test-set metrics and Art-Any subset metrics. The former are computed over all edges or triples in the test set and reflect the overall predictive performance on the complete ARTNET graph. The latter are computed only on test samples that contain at least one art node, providing a more focused evaluation of model performance in art-related relation prediction. Therefore, the Art-Any metrics more directly reflect how well the model captures art nodes and their associated semantic relations.

For link prediction, most models perform better on the Art-Any subset than on the corresponding full test set, which indicates that models obtain stronger semantic support when predicting relations involving art nodes. For example, RoBERTa + ResNet-50 achieves a full-set AUC-PR of 0.1457, while its Art-Any AUC-PR increases to 0.4186. Its full-set Hit@10 is 0.7606, while its Art-Any Hit@10 increases to 0.8992. This gap suggests that when nodes contain richer textual and visual information, multimodal representations provide more effective discriminative signals, improving both real-edge identification and candidate-relation ranking. In this sense, the Art-Any results provide more direct experimental support for the effectiveness of multimodal information in art-related relation modeling.

The results across different backbone combinations further show that text semantic modeling, visual representation quality, and cross-modal fusion stability all affect link prediction performance. RoBERTa + ResNet-50 achieves the best results on both full-set and Art-Any metrics, suggesting that a strong text encoder combined with a stable pretrained visual backbone is better suited to multimodal relation reasoning in ARTNET. ALBERT + Inception-ResNet-v2 performs well on full-set AUC-PR, but its Hit@10 and Art-Any Hit@10 are clearly lower than those of RoBERTa + ResNet-50. This indicates that it has certain global discriminative ability, but remains limited in ranking true relations near the top of the candidate list. BERT + DINOv2 and BERT + Faster R-CNN ResNet50 FPN V2 show relatively strong performance on the Art-Any subset, indicating that self-supervised visual features and region-level object features can provide useful complementary signals for relation prediction involving art nodes. However, their full-set performance remains constrained by text representation capacity and cross-modal coordination. In contrast, BERT + VGG-19 and the trainable BERT + ResNet-18 setting show weaker overall performance, suggesting that traditional visual features or direct end-to-end training of a lightweight visual backbone may not fit the current large-scale graph link prediction setting. A possible reason is that the supervision signal mainly comes from edge-existence prediction rather than a direct visual classification objective. As a result, the visual branch is more vulnerable to image noise, missing modalities, and sparse edge supervision, whereas frozen pretrained visual backbones provide more stable image-semantic representations.

For edge-type prediction, the pure KGE results show that model performance depends not only on multimodal entity representations, but also strongly on the relation scoring function. ComplEx performs better overall on MRR, Hit@10, and AUC-PR, indicating that its bilinear formulation in complex space is more suitable for representing the complex, diverse, and potentially asymmetric relation semantics in ARTNET. Under the ComplEx scorer, ALBERT + Inception-ResNet-v2 achieves the highest MRR and Hit@10, suggesting stronger performance in true-relation ranking and top-ranked retrieval. RoBERTa + ResNet-50 has slightly lower ranking metrics, but achieves the highest AUC-PR and Art-Any AUC-PR, indicating more stable performance in triple classification and art-related relation identification. In comparison, TransE is weaker than ComplEx in overall ranking performance, but RoBERTa + ResNet-50 remains the most stable combination within the TransE group. This suggests that strong text representations and stable visual features retain a degree of robustness across different scoring functions.

The Art-Any subset reveals a more nuanced pattern in edge-type prediction. Most models achieve substantially higher Art-Any MRR than full-set MRR, indicating that multimodal information helps improve the ranking quality of art-related relations. However, Art-Any AUC-PR is not always consistently higher than full-set AUC-PR. This shows that art-related samples contain richer multimodal evidence, but their relation semantics are also more complex, so the classification difficulty is not necessarily lower. Therefore, the Art-Any metrics should not be interpreted simply as an easier test subset. Instead, they provide an evaluation perspective that is closer to the core application scenario of the art domain.

The GraphKGE experiments further demonstrate the effect of graph-structural information on edge-type prediction. Unlike pure KGE, which relies only on multimodal entity representations and relation embeddings, GraphKGE introduces a GraphSAGE or R-GCN graph encoder before scoring, allowing entity representations to absorb local neighborhood structure. The results show that R-GCN clearly outperforms GraphSAGE on MRR and Art-Any MRR, especially under the TransE scorer. For example, TransE + ALBERT + Inception-ResNet-v2 + R-GCN achieves the highest MRR and Art-Any MRR among all GraphKGE models, indicating that relation-aware graph message passing can substantially strengthen relation-ranking ability. This result is consistent with the nature of edge-type prediction. Compared with aggregating general topological neighbors, R-GCN distinguishes information propagation paths associated with different relation types, making it more suitable for modeling the semantics of typed edges.

Nevertheless, GraphSAGE remains competitive on some AUC-PR metrics, suggesting that general topological structure is still useful for triple classification, although its ability to distinguish fine-grained relation types is weaker than that of R-GCN. The GraphKGE results also show that graph propagation does not bring consistent gains across all metrics. Model performance is jointly affected by the scoring function, multimodal entity representation, and graph encoder design. Therefore, the value of GraphKGE lies not only in improving certain metrics, but also in providing a comparative perspective for analyzing the different roles of content representation, general topological structure, and relation-aware structural information in edge-type prediction.

Overall, the experimental results show that multimodal information and graph-structural information play practical roles in art-related relation modeling in ARTNET. The link prediction results show that the clear improvements on the Art-Any subset indicate that textual and visual modalities provide more effective semantic signals for judging edge existence around art nodes. The pure KGE results show that different multimodal entity representations and relation scoring functions substantially affect edge-type ranking and classification. The GraphKGE results further show that relation-aware graph propagation can enhance relation-semantic modeling. These findings indicate that multimodal content, graph-structural context, and relation-type information play complementary roles in modeling the ARTNET art knowledge graph. They also show that the proposed evaluation framework can comprehensively assess model performance at both the full-graph prediction level and the art-related relation prediction level.

5.2 Limitations

Although the paper evaluates model performance using both full-test-set metrics and Art-Any subset metrics, the current results cannot be directly interpreted as strict causal evidence for the contribution of each modality. The Art-Any subset provides a more focused view of model performance in art-related relation prediction, but its results may still be affected by relation-type distributions, node-degree distributions, negative-sample difficulty, and entity semantic density. Therefore, the Art-Any results show that the models achieve stronger predictive performance on art-related samples and provide experimental support for the role of multimodal information in art relation modeling. However, further quantification of the individual contributions of text, image, and graph-structural information requires future ablation studies with text-only, image-only, structure-only, and multimodal settings, together with a small number of case studies to examine the semantic quality of model predictions.

In terms of model design, the TransE, ComplEx, GraphSAGE, and R-GCN methods used in this paper are classical and stable knowledge graph modeling approaches. They are suitable as initial baselines on ARTNET, but they also limit the upper bound of model expressiveness. TransE has limited capacity to capture complex relation patterns, GraphSAGE does not explicitly distinguish relation types, and R-GCN introduces relation-aware message passing but incurs higher computational cost on large-scale graphs. The focus of this paper is to validate ARTNET as a usable multimodal art knowledge graph infrastructure and to establish basic experimental frameworks for link prediction, pure KGE, and GraphKGE. Therefore, the paper does not further incorporate more complex knowledge graph pretraining models, cross-modal attention mechanisms, or large-scale vision-language models. Future work can build on the current experimental setting to explore stronger multimodal fusion methods and relation modeling structures.

In terms of system scale, ARTNET contains graph structure, textual descriptions, and image information, which makes multimodal training substantially more expensive than single-modal knowledge graph modeling. This cost is especially evident in GraphKGE, where the model requires neighborhood sampling and graph message passing during both training and inference, further increasing computational and memory overhead. To control experimental cost, the paper restricts batch size, the number of sampled neighbors, the maximum number of subgraph nodes, and the number of samples used in some evaluations. These restrictions may prevent the model from fully exploiting all structural information in the complete graph. For some art relations that depend on long-range semantic paths or cross-region context, the current local-subgraph sampling strategy may still be insufficient.

Finally, the experimental conclusions are mainly based on the current ARTNET dataset and the current data split, and their generalization ability still needs further validation on more datasets and task settings. Art-domain data usually has strong multimodal characteristics, where image content and textual descriptions are highly valuable for relation modeling, so multimodal methods are more likely to show advantages. However, on datasets with more severe image missingness, lower text quality, or more homogeneous relation types, similar methods may show weaker performance. Therefore, the results of this paper are better understood as an initial evaluation benchmark on the ARTNET dataset. Future work still needs to examine model stability and generalization ability on more art datasets, cross-domain knowledge graphs, and more fine-grained task settings.

6 Conclusion

This paper proposes ARTNET, a multimodal knowledge graph infrastructure for relation modeling in the art domain. ARTNET organizes art-related heterogeneous entities, typed relations, textual descriptions, and visual resources within a unified graph framework, and uses graph partitioning to reduce memory pressure and computational cost during large-scale graph processing. Based on this infrastructure, the paper defines and evaluates two core tasks: multimodal link prediction, which determines whether a connection should exist between two entities, and multimodal edge-type scoring, which predicts the semantic relation type between entities. Through these two tasks, the paper provides a systematic experimental framework for evaluating whether textual information, visual content, and graph-structural information can jointly support art-related relation reasoning.

The experimental results show that multimodal entity representations provide effective signals for relation prediction on ARTNET. In the link prediction experiments, stronger models achieve high performance not only on the full test set, but also on the Art-Any subset. Since the Art-Any subset contains candidate node pairs in which at least one node is an art-related entity, this result suggests that the models do not rely only on general graph connectivity patterns, but also capture semantic and visual information in art-related relations to some extent. The edge-type scoring experiments further show that model performance is affected by both the multimodal backbone and the relation scoring function. Under the pure KGE setting, ComplEx is generally more stable than TransE, indicating that stronger relation modeling capacity helps represent the diverse and potentially asymmetric relation semantics in ARTNET. Under the GraphKGE setting, the introduction of GraphSAGE and R-GCN further demonstrates the potential value of structural context for edge-type modeling, although the improvement is not consistent across all metrics and scorers. This suggests that graph-structural information is useful, but its effect depends on how relation information, neighborhood aggregation, and multimodal entity features are combined.

Overall, the experiments support the central goal of this paper: ARTNET is not only a large-scale art knowledge graph, but also a reusable experimental infrastructure for multimodal relation modeling. Full-test-set metrics reflect the predictive ability of models over the overall graph structure, while Art-Any metrics provide a more targeted evaluation of model performance in art-related relation prediction. By reporting both types of metrics, the paper distinguishes general graph modeling ability from the ability to model relations involving art-related entities. The results provide initial evidence that multimodal information has a positive role in art-related relation modeling, and establish baseline references for future work on link prediction, KGE relation scoring, and graph-enhanced relation modeling on ARTNET.

Future work can extend this study in several directions. First, stronger vision-language alignment methods can be incorporated, such as contrastive multimodal encoders or vision-language models adapted to the art domain, to improve the consistency between textual descriptions and visual features. Second, more powerful relation-aware graph encoders can be explored, especially for heterogeneous graphs with many relation types and imbalanced relation distributions, to improve the modeling of complex relation patterns. Third, the evaluation protocol can be further strengthened by introducing harder negative sampling strategies, more fine-grained art-subset evaluations, and qualitative case studies, so that model predictions can be assessed not only through ranking metrics but also from the perspective of art-historical semantics. Finally, ARTNET can be expanded with richer artwork metadata, higher-quality visual resources, and more downstream tasks, such as artwork retrieval, visual subject relation prediction, and art knowledge discovery. These extensions can further enhance the research value of ARTNET as a scalable multimodal knowledge infrastructure for the art domain.

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